

What a domain prior can and cannot replace

Cajetan Songwae

songwae2000@gmail.com

TGDC case study, research track

Every figure below is printed by `python run_all.py`.

Question

Labs generate enterprise data from procedurally built companies: an org chart, a taxonomy of work, and hand-written rules for how that work behaves. If domain reasoning can supply the behaviour, a generator can be built for a client without their records.

This paper tests whether it can, for one concrete property of operational work: how long a task takes.

Hypothesis

H. Domain priors reasoned from the nature of the work capture a substantial share of what real operational records provide for predicting task duration.

Recorded before evaluation: a prior written without sight of the target data should reach an AUC of 0.60 to 0.72. Below 0.55 would mean domain reasoning supplies essentially nothing.

Why it matters. If H holds, a taxonomy is enough to build a generator. If it fails, the behaviour has to come from the client's own data.

Method

Data. Public 311 service request feeds from four cities. These are real operational records, written by the organisations that do the work. Two complete calendar weeks from each, the weeks of 4 May and 8 June 2026.

city	train	test	median resolution	categories
New York	72,206	78,296	5.4h	132
Chicago	17,291	38,671	118.0h	97
Austin	5,968	6,029	23.8h	117
San Francisco	17,723	17,007	15.1h	37

Tickets still open at the pull date are kept and treated as slow, since all are months past any threshold used. Chicago closes most of its feed within a second of creation, so those informational records are excluded.

Task. At ticket opening, predict whether it will take longer than the median to resolve. Logistic regression on one-hot categoricals, so the learner is held constant. Recovered skill is the share of the achievable advantage over chance that an arm reaches: $(\text{arm} - 0.5) / (\text{ceiling} - 0.5)$.

The ceiling is the prior's own information. A prior reasons over the service category and the owning department. Fitting the conditional rate directly from those same two columns gives the best any rule of that shape could do. Comparing against it separates a weak rule from weak features.

Blind authorship. Rules were written from each city's service type and department vocabulary alone. The taxonomy was pulled with the query restricted to names and row counts, so no duration, rate or outcome was available. Rules were written, committed with a recorded prediction, and only then evaluated.

Two authors. The largest threat to a result like this is that it describes one person's reasoning. Each city was given two independent authors under the same blind condition: the first was the experimenter, the second a language model that saw only the taxonomy and was instructed not to look anything up.

Results

What a blind prior recovers

city	author	AUC [95% CI]	ceiling	recovered	vs chance
Austin	human	0.494 [0.481, 0.508]	0.883	−1%	indistinguishable
Austin	independent	0.611 [0.598, 0.626]	0.883	29%	above
San Francisco	human	0.676 [0.670, 0.684]	0.886	46%	above
San Francisco	independent	0.777 [0.770, 0.784]	0.886	72%	above

Intervals are 1,000-resample bootstraps. The null baseline, always predicting the majority class, scores 0.500 by construction.

H cannot be answered yes or no. A blind prior recovers 72% of the achievable skill in San Francisco and nothing at all in Austin. Testing one city would have produced a confident conclusion in either direction, and the direction would have been an artefact of the city.

Two effects sit inside that spread, and they differ in size. **Author** is consistent and modest: the independent author beats the experimenter by 0.117 in Austin and 0.101 in San Francisco, the same direction and roughly the same magnitude twice. **City** is much larger: the same author moves from 29% to 72%.

Three predictions, recorded before evaluation

arm	predicted	observed	inside band
Austin, human	0.60 to 0.72	0.494	no
Austin, independent	0.52 to 0.62	0.611	yes
San Francisco, human	0.55 to 0.68	0.676	yes

The first band was set before any result existed and was wrong. The later two were set knowing the Austin outcome and were right, which is a weaker achievement and is reported as such.

One category family explains the gap between the cities

subset	n	prior	ceiling	recovered
Austin, all tickets	6,029	0.611	0.883	29%
Austin, excluding waste collection	4,008	0.766	0.881	70%
San Francisco, all tickets	17,007	0.777	0.886	72%

Remove one family of work from Austin and its blind prior recovers 70%, which is San Francisco’s 72%. The cities were never different. One category family was.

Waste collection is 34% of Austin’s volume and 70% of it runs slow. The prior calls it fast, mean prediction 0.37, and the reason is worth stating plainly. A missed-collection ticket stays open until the next scheduled route, which is a week away. The work is trivially routine. The ticket is not. Nothing in the phrase ARR - Compost reveals that.

San Francisco has no equivalent. Its two largest categories, street cleaning and parking enforcement, are 57% of volume, and both behave the way the work suggests.

city	categories	top 5 share	volume in wrong-direction categories
Austin	117	40%	35%
San Francisco	37	78%	17%

The finding

A blind domain prior recovers roughly 70% of the achievable skill on work whose duration follows from the nature of the job, in both cities and for both authors. It recovers nothing on work whose duration is set by an administrative cycle the job description does not reveal. What determines its value is how much of a client’s volume sits in the second group, and that is not knowable from the taxonomy.

What having the rates is worth

prior	tested on	AUC [95% CI]	ceiling	recovered
New York	New York	0.840 [0.837, 0.843]	0.918	81%
New York, carried across	Chicago	0.496 [0.492, 0.499]	0.717	−2%

Rules written with a city’s conditional rates in hand reach 81% there, above the 72% a blind author reached in San Francisco but not by much. The same rules carried to another city fall entirely below chance. Their interval excludes 0.5 from beneath, so they are not merely uninformative but actively misleading. The department rules match 0.0% of Chicago rows, being New York acronyms.

Distribution fidelity does not detect any of this

generator	AUC	recovered	records that cannot occur
real records (ceiling)	0.918	100%	0.0%
marginal only	0.466	−8%	70.5%
marginals + pairwise	0.917	100%	0.0%
empirical joint	0.917	100%	0.0%

Matching marginals and pairwise dependence, the bar SDV’s Quality Score and the NIST differential-privacy winners optimise, recovers everything on the New York task and invents no impossible records. A second recorded prediction, that impossible records would explain downstream loss, therefore failed at the fidelity order that matters.

Limits

Four cities, one domain, one task family, one model class. Municipal services may be unusually institution-bound. Two authors, and one of them is the experimenter. The independent author is a language model, which is a real second author but not a domain practitioner. A retired public works manager might do better than either.

The waste-collection result is a single family in a single city. It is an explanation consistent with the numbers rather than a tested mechanism, and the way to test it is to predict in advance, in a fifth city, which families will break a prior and check.

The San Francisco human prior was written knowing what Austin had shown. Its rules are blind to San Francisco’s rates but its author is not blind to the general lesson, so that arm tests transfer of a lesson rather than reasoning from scratch.

Chicago’s June week ran roughly twice the volume of its May week, so the carried arm compares two operating conditions as well as two cities.

New York carries known artefacts, including one department that closes 89.7% of its tickets at exactly midnight, so its recorded durations are administrative.

That low-order fidelity fails to imply downstream utility is established. The ladder is included because it shows a distribution-level check cannot see the effect this paper measures.

What it would take to answer the general question

Predict the failure families before looking. The waste-collection explanation implies that any category whose clock is an administrative cycle rather than the work itself will break a blind prior. That is checkable: name the categories in a fifth city from the taxonomy, record the list, then measure.

More authors, including practitioners, would separate the procedure from the person further than two can.

The procedural point stands regardless. A prior can only be evaluated as a prior once per dataset, before anyone has seen the outcomes, and the recorded prediction has to come first. Of the three predictions here, the only one made in genuine ignorance was the one that turned out wrong.